Database Management System LAB
Spring Semester 2024-25
Lab Assignment 4

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Table Creation

Foreign Keys will be added later on after creating all tables

```
mysql> create table Employees
    -> (Employee_id int PRIMARY KEY,
    -> First_name Varchar(20),
    -> Last_name Varchar(20),
    -> Email Varchar(30),
    -> Phone_number int,
    -> Hire_date DATE,
    -> Job_id int,
    -> Salary int,
    -> Commission_pct int,
    -> Manager_id int,
    -> Department_id int);
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> load data local infile 'Employee.txt' into table Employees;
Query OK, 10 rows affected (0.06 sec)
Records: 10  Deleted: 0  Skipped: 0  Warnings: 0

mysql> select * from Employees;
+-------------+------------+-----------+----------------+--------------+------------+--------+--------+----------------+------------+---------------+
| Employee_id | First_name | Last_name | Email          | Phone_number | Hire_date  | Job_id | Salary | Commission_pct | Manager_id | Department_id |
+-------------+------------+-----------+----------------+--------------+------------+--------+--------+----------------+------------+---------------+
|           1 | John       | Smith     | john@google.com |    123456789 | 1965-01-09 |      1 |  30000 |              2 |          1 |            10 |
|           2 | Frank      | Wong      | frank@gmail.com |    333445555 | 1955-12-08 |      4 |  40000 |              3 |          1 |            10 |
|           3 | Alicia     | Zelaya    | alici@gmail.com |    999887777 | 1968-01-19 |      4 |  25000 |              5 |          4 |            20 |
|           4 | Jenni      | Wall      | jenni@gmail.com |    987654321 | 1941-06-20 |      1 |  43000 |              7 |          4 |            20 |
|           5 | Ramesh     | Narayan   | rames@gmail.com |    666884444 | 1962-09-15 |      2 |  38000 |             11 |          1 |            10 |
|           6 | Joyce      | English   | joyce@gmail.com |    453453453 | 1972-07-31 |      2 |  25000 |             13 |          1 |            10 |
|           7 | Ahmad      | Jabbar    | ahmad@gmail.com |    987987987 | 1969-03-29 |      5 |  25000 |             17 |          1 |            10 |
|           8 | James      | Borg      | james@gmail.com |    888665555 | 1937-11-10 |      3 |  75000 |             19 |          9 |            30 |
|           9 | Thomas     | Ding      | thoma@gmail.com |    768768768 | 1947-08-15 |      5 |  35000 |             23 |          9 |            30 |
|          10 | Ding       | Liren     | dingg@gmail.com |    123454321 | 2004-08-14 |      3 |  60000 |             29 |          1 |            10 |
+-------------+------------+-----------+----------------+--------------+------------+--------+--------+----------------+------------+---------------+
10 rows in set (0.00 sec)
```

Open Employee.txt
~/Documents/DBMS/Assignment_4

● 1 Clerk Employee.txt

```
1       John    Smith   john@google.com 123456789       1965-01-09      1       30000   2       1       10
2       Frank   Wong    frank@gmail.com 333445555       1955-12-08      4       40000   3       1       10
3       Alicia  Zelaya  alici@gmail.com 999887777       1968-01-19      4       25000   5       4       20
4       Jenni   Wall    jenni@gmail.com 987654321       1941-06-20      1       43000   7       4       20
5       Ramesh  Narayan rames@gmail.com 666884444       1962-09-15      2       38000   11      1       10
6       Joyce   English joyce@gmail.com 453453453       1972-07-31      2       25000   13      1       10
7       Ahmad   Jabbar  ahmad@gmail.com 987987987       1969-03-29      5       25000   17      1       10
8       James   Borg    james@gmail.com 888665555       1937-11-10      3       75000   19      9       30
9       Thomas  Ding    thoma@gmail.com 768768768       1947-08-15      5       35000   23      9       30
10      Ding    Liren   dingg@gmail.com 123454321       2004-08-14      3       60000   29      1       10
```

```
mysql> create table Departments
    -> (Department_id int PRIMARY KEY,
    -> Department_name Varchar(20),
    -> Manager_id int,
    -> Location_id int);
Query OK, 0 rows affected (0.10 sec)

mysql> load data local infile 'Department.txt' into table Departments;
Query OK, 3 rows affected (0.07 sec)
Records: 3  Deleted: 0  Skipped: 0  Warnings: 0

mysql> select * from Departments;
+---------------+-----------------+------------+-------------+
| Department_id | Department_name | Manager_id | Location_id |
+---------------+-----------------+------------+-------------+
|            10 | Research        |          1 |           1 |
|            20 | Administration  |          2 |           2 |
|            30 | Headquarters    |          3 |           3 |
+---------------+-----------------+------------+-------------+
3 rows in set (0.00 sec)
```

Open Department.txt
~/Documents/DBMS/Assignment_4

```
10      Research        1       1
20      Administration  2       2
30      Headquarters    3       3
```

```
mysql> create table Jobs
    -> (Job_id int PRIMARY KEY,
    -> Job_Title Varchar(20),
    -> Min_Salary int,
    -> Max_Salary int);
Query OK, 0 rows affected (0.05 sec)

mysql> load data local infile 'Job.txt' into table Jobs;
Query OK, 5 rows affected (0.00 sec)
Records: 5  Deleted: 0  Skipped: 0  Warnings: 0

mysql> select * from Jobs;
+--------+------------+------------+------------+
| Job_id | Job_Title  | Min_Salary | Max_Salary |
+--------+------------+------------+------------+
|      1 | Managerial |      30000 |      43000 |
|      2 | Clerk      |      25000 |      38000 |
|      3 | Drafter    |      60000 |      75000 |
|      4 | Blacksmith |      25000 |      40000 |
|      5 | Goldsmith  |      25000 |      35000 |
+--------+------------+------------+------------+
5 rows in set (0.00 sec)
```

Open

Job.txt
~/Documents/DBMS/Assignment_4

```
1       Managerial      30000   43000
2       Clerk   25000   38000
3       Drafter 60000   75000
4       Blacksmith      25000   40000
5       Goldsmith       25000   35000
```

```
mysql> create table Job_History
    -> (Employee_id int,
    -> Start_date DATE,
    -> End_Date DATE,
    -> Job_id int,
    -> Department_id int,
    -> PRIMARY KEY(Employee_id,Start_date));
Query OK, 0 rows affected (0.10 sec)

mysql> load data local infile 'Job_History.txt' into table Job_History;
Query OK, 10 rows affected (0.01 sec)
Records: 10  Deleted: 0  Skipped: 0  Warnings: 0

mysql> select * from Job_History;
+-------------+------------+------------+--------+---------------+
| Employee_id | Start_date | End_Date   | Job_id | Department_id |
+-------------+------------+------------+--------+---------------+
|           1 | 1965-01-09 | 2004-01-07 |      1 |            10 |
|           2 | 1955-12-08 | 2005-03-18 |      4 |            10 |
|           3 | 1968-01-19 | 2006-04-06 |      4 |            20 |
|           4 | 1941-06-20 | 2007-05-15 |      1 |            20 |
|           5 | 1962-09-15 | 2008-06-23 |      2 |            10 |
|           6 | 1972-07-31 | 2009-07-12 |      2 |            10 |
|           7 | 1969-03-29 | 2009-08-14 |      5 |            10 |
|           8 | 1937-11-10 | 2010-09-15 |      3 |            30 |
|           9 | 1947-08-15 | 2014-10-13 |      5 |            30 |
|          10 | 2004-08-14 | 2024-02-03 |      3 |            10 |
+-------------+------------+------------+--------+---------------+
10 rows in set (0.00 sec)
```

Open

Job_History.txt
~/Documents/DBMS/Assignment_4

Employee.txt | Job_History.txt | Job.txt

```
1       1965-01-09      2004-01-07      1       10
2       1955-12-08      2005-03-18      4       10
3       1968-01-19      2006-04-06      4       20
4       1941-06-20      2007-05-15      1       20
5       1962-09-15      2008-06-23      2       10
6       1972-07-31      2009-07-12      2       10
7       1969-03-29      2009-08-14      5       10
8       1937-11-10      2010-09-15      3       30
9       1947-08-15      2014-10-13      5       30
10      2004-08-14      2024-02-03      3       10
```

```
mysql> create table Locations
    -> (Location_id int PRIMARY KEY,
    -> Street_Address Varchar(20),
    -> Postal_code int,
    -> City Varchar(20),
    -> State_Province Varchar(20),
    -> Country_id int);
Query OK, 0 rows affected (0.04 sec)

mysql> load data local infile 'Location.txt' into table Locations;
Query OK, 3 rows affected (0.01 sec)
Records: 3  Deleted: 0  Skipped: 0  Warnings: 0

mysql> select * from Locations;
+-------------+----------------+------------+---------+----------------+------------+
| Location_id | Street_Address | Postal_code | City    | State_Province | Country_id |
+-------------+----------------+------------+---------+----------------+------------+
|           1 | Wall Street    |      710345 | Noida   | Delhi          |          1 |
|           2 | Shibuya        |      109441 | Kyoto   | Tokyo          |          2 |
|           3 | MG Road        |      719332 | Gangtok | Sikkim         |          1 |
+-------------+----------------+------------+---------+----------------+------------+
3 rows in set (0.00 sec)
```

Open

Location.txt
~/Documents/DBMS/Assignment_4

Department.txt | Location.txt

```
1       Wall Street     710345  Noida   Delhi   1
2       Shibuya 109441  Kyoto   Tokyo   2
3       MG Road 719332  Gangtok Sikkim  1
```

```
mysql> create table Countries
    -> (Country_id int PRIMARY KEY,
    -> Country_name Varchar(20),
    -> Region_id int);
Query OK, 0 rows affected (0.10 sec)

mysql> load data local infile 'Country.txt' into table Countries;
Query OK, 2 rows affected (0.01 sec)
Records: 2  Deleted: 0  Skipped: 0  Warnings: 0

mysql> select * from Countries;
+------------+--------------+-----------+
| Country_id | Country_name | Region_id |
+------------+--------------+-----------+
|          1 | India        |         1 |
|          2 | Japan        |         2 |
+------------+--------------+-----------+
2 rows in set (0.00 sec)
```

Open

Country.txt
~/Documents/DBMS/Assignment_4

Location.txt | Country.txt

```
1       India   1
2       Japan   2
```

```
mysql> create table Regions
    -> (Region_id int PRIMARY KEY,
    -> Region_name Varchar(20));
Query OK, 0 rows affected (0.05 sec)

mysql> load data local infile 'Region.txt' into table Regions;
Query OK, 2 rows affected (0.01 sec)
Records: 2  Deleted: 0  Skipped: 0  Warnings: 0

mysql> select * from Regions;
+-----------+------------+
| Region_id | Region_name |
+-----------+------------+
|         1 | Asia West   |
|         2 | Asia East   |
+-----------+------------+
2 rows in set (0.01 sec)
```

Open

Region.txt
~/Documents/DBMS/Assignment_4

Country.txt | Region.txt

```
1       Asia West
2       Asia East
```

Adding Foreign Key Constraints

```
mysql> alter table Employees
    -> Add Constraint Foreign Key (Department_id) References Departments(Department_id)
    -> on Delete CASCADE;
Query OK, 10 rows affected (0.16 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> alter table Employees
    -> Add Constraint Foreign Key (Manager_id) References Employees(Employee_id)
    -> on Delete CASCADE;
Query OK, 10 rows affected (0.18 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> alter table Employees
    -> Add Constraint Foreign Key (Job_id) References Jobs(Job_id)
    -> on Delete CASCADE;
Query OK, 10 rows affected (0.20 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> alter table Departments
    -> Add Constraint Foreign Key (Manager_id) References Employees(Employee_id)
    -> on Delete CASCADE;
Query OK, 3 rows affected (0.11 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> alter table Job_History
    -> Add Constraint Foreign Key (Employee_id) References Employees(Employee_id)
    -> on Delete CASCADE;
Query OK, 10 rows affected (0.10 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> alter table Job_History
    -> Add Constraint Foreign Key (Department_id) References Departments(Department_id)
    -> on Delete CASCADE;
Query OK, 10 rows affected (0.19 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> alter table Job_History
    -> Add Constraint Foreign Key (Job_id) References Jobs(Job_id)
    -> on Delete CASCADE;
Query OK, 10 rows affected (0.19 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> alter table Locations
    -> add Constraint Foreign Key (Country_id) References Countries(Country_id)
    -> on Delete CASCADE;
Query OK, 3 rows affected (0.24 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> alter table Departments
    -> add Constraint Foreign Key (Location_id) References Locations(Location_id)
    -> on Delete CASCADE;
Query OK, 3 rows affected (0.13 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> alter table Countries
    -> add Constraint Foreign Key (Region_id) References Regions(Region_id)
    -> on Delete CASCADE;
Query OK, 2 rows affected (0.11 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

Ans – 1)

```
mysql> select Concat(First_name," ",Last_name) as Emp_Name,Salary as Monthly_Pay,Salary*12 as Annual_Pay,d.Department_name
    -> from Employees as e
    -> Join Departments as d on d.Department_id = e.Department_id
    -> Order by Salary DESC;
+----------------+-------------+------------+-----------------+
| Emp_Name       | Monthly_Pay | Annual_Pay | Department_name |
+----------------+-------------+------------+-----------------+
| James Borg     |       75000 |     900000 | Headquarters    |
| Ding Liren     |       60000 |     720000 | Research        |
| Jenni Wall     |       43000 |     516000 | Administration  |
| Frank Wong     |       40000 |     480000 | Research        |
| Ramesh Narayan |       38000 |     456000 | Research        |
| Thomas Ding    |       35000 |     420000 | Headquarters    |
| John Smith     |       30000 |     360000 | Research        |
| Joyce English  |       25000 |     300000 | Research        |
| Ahmad Jabbar   |       25000 |     300000 | Research        |
| Alicia Zelaya  |       25000 |     300000 | Administration  |
+----------------+-------------+------------+-----------------+
10 rows in set (0.00 sec)
```

Ans – 2)

```
mysql> select m.First_name as Manager_Name,COUNT(*) as No_of_Employees
    -> from Employees as e
    -> Join Departments as d on (d.Department_name = 'Research' and d.Department_id = e.Department_id)
    -> Join Employees as m on (e.Manager_id = m.Employee_id)
    -> Group by m.First_name;
+--------------+-----------------+
| Manager_Name | No_of_Employees |
+--------------+-----------------+
| John         |               6 |
+--------------+-----------------+
1 row in set (0.01 sec)
```

Ans – 3)

```
mysql> select Concat(First_name," ",Last_name) as Emp_Name,Salary as Monthly_Pay,Salary*12 as Annual_Pay,d.Department_name
    -> from Employees as e
    -> Join Departments as d on d.Department_id = e.Department_id
    -> Order by (Salary*(1+Commission_pct/100)) DESC;
+----------------+-------------+------------+-----------------+
| Emp_Name       | Monthly_Pay | Annual_Pay | Department_name |
+----------------+-------------+------------+-----------------+
| James Borg     |       75000 |     900000 | Headquarters    |
| Ding Liren     |       60000 |     720000 | Research        |
| Jenni Wall     |       43000 |     516000 | Administration  |
| Thomas Ding    |       35000 |     420000 | Headquarters    |
| Ramesh Narayan |       38000 |     456000 | Research        |
| Frank Wong     |       40000 |     480000 | Research        |
| John Smith     |       30000 |     360000 | Research        |
| Ahmad Jabbar   |       25000 |     300000 | Research        |
| Joyce English  |       25000 |     300000 | Research        |
| Alicia Zelaya  |       25000 |     300000 | Administration  |
+----------------+-------------+------------+-----------------+
10 rows in set (0.00 sec)
```

Ans – 4)

```
mysql> select Concat(First_name," ",Last_name) as Emp_Name,TimeStampDiff(YEAR,Hire_Date,CURDATE()) as Experience
    -> from Employees
    -> Order by Experience DESC;
+----------------+------------+
| Emp_Name       | Experience |
+----------------+------------+
| James Borg     |         87 |
| Jenni Wall     |         83 |
| Thomas Ding    |         77 |
| Frank Wong     |         69 |
| Ramesh Narayan |         62 |
| John Smith     |         60 |
| Alicia Zelaya  |         57 |
| Ahmad Jabbar   |         55 |
| Joyce English  |         52 |
| Ding Liren     |         20 |
+----------------+------------+
10 rows in set (0.00 sec)
```

Ans – 5) Note that here There is no DeptNo = 90 present in this table
Alternate query for DeptNo = 30 (as it is present in table)

```
  GNU nano 6.2                                                    /tmp/sql0ajDt7
update Employees
set Salary = Salary * 1.1 where Department_id = 90;

Select (Salary/(1+0.1*(Department_id = 90))) as old_Salary,Salary as new_Salary
from Employees;

# Note that there are no changes in the new and old Salary in the table because there are no entries made after DeptNo=90
# Hence just for demonstration, I am updating DeptNo = 30 as it is present in the table and to show the changes made in
# the Salaries are actually correct.

update Employees
set Salary = Salary * 1.1 where Department_id = 30;

Select (Salary/(1+0.1*(Department_id = 30))) as old_Salary,Salary as new_Salary
from Employees;
```

```
mysql> \e;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0

+------------+------------+
| old_Salary | new_Salary |
+------------+------------+
| 30000.0000 |      30000 |
| 40000.0000 |      40000 |
| 25000.0000 |      25000 |
| 43000.0000 |      43000 |
| 38000.0000 |      38000 |
| 25000.0000 |      25000 |
| 25000.0000 |      25000 |
| 75000.0000 |      75000 |
| 35000.0000 |      35000 |
| 60000.0000 |      60000 |
+------------+------------+
10 rows in set (0.00 sec)

Query OK, 2 rows affected (0.00 sec)
Rows matched: 2  Changed: 2  Warnings: 0

+------------+------------+
| old_Salary | new_Salary |
+------------+------------+
| 30000.0000 |      30000 |
| 40000.0000 |      40000 |
| 25000.0000 |      25000 |
| 43000.0000 |      43000 |
| 38000.0000 |      38000 |
| 25000.0000 |      25000 |
| 25000.0000 |      25000 |
| 75000.0000 |      82500 |
| 35000.0000 |      38500 |
| 60000.0000 |      60000 |
+------------+------------+
10 rows in set (0.00 sec)
```

Ans – 6)

```
  GNU nano 6.2                                          /tmp/sqlwIGBEz
select d.Department_id,Department_name,p.Salary

from Departments as d

Join (Select Department_id,sum(Salary) as Salary
from Employees Group by Department_id) as p on p.Department_id = d.Department_id

where p.Salary > 100000;
mysql> \e;
+---------------+-----------------+--------+
| Department_id | Department_name | Salary |
+---------------+-----------------+--------+
|            10 | Research        | 218000 |
|            30 | Headquarters    | 121000 |
+---------------+-----------------+--------+
2 rows in set (0.01 sec)
```

Ans – 7)

```
  GNU nano 6.2                                        /tmp/sqlC5ViSO *
create view min_max as

select e.Job_id,mini.Name as Minimum_earner,maxi.Name as Maximum_earner
from Employees as e

join (select e1.Job_id as job_id,concat(e1.First_name," ",e1.Last_name) as Name
from Employees as e1
Left Join Employees as e2 on (e1.Job_id = e2.Job_id and e1.Salary > e2.Salary)
where e2.Salary is NULL) as mini on mini.Job_id = e.Job_id

join (select e1.Job_id as job_id,concat(e1.First_name," ",e1.Last_name) as Name
from Employees as e1
Left Join Employees as e2 on (e1.Job_id = e2.Job_id and e1.Salary < e2.Salary)
where e2.Salary is NULL) as maxi on maxi.Job_id = e.Job_id

Group by Job_id;

select * from min_max;
```

```
mysql> \e;
Query OK, 0 rows affected (0.03 sec)

+--------+----------------+----------------+
| Job_id | Minimum_earner | Maximum_earner |
+--------+----------------+----------------+
|      4 | Alicia Zelaya  | Frank Wong     |
|      1 | John Smith     | Jenni Wall     |
|      2 | Joyce English  | Ramesh Narayan |
|      3 | Ding Liren     | James Borg     |
|      5 | Ahmad Jabbar   | Thomas Ding    |
+--------+----------------+----------------+
5 rows in set (0.04 sec)
```

```
mysql> insert into Employees
    -> Values (11,'New','Maximum','new@gmail.com',742589533,'1942-03-29',2,95000,10,4,20);
Query OK, 1 row affected (0.02 sec)

mysql> select * from min_max;
+--------+----------------+----------------+
| Job_id | Minimum_earner | Maximum_earner |
+--------+----------------+----------------+
|      4 | Alicia Zelaya  | Frank Wong     |
|      1 | John Smith     | Jenni Wall     |
|      3 | Ding Liren     | James Borg     |
|      5 | Ahmad Jabbar   | Thomas Ding    |
|      2 | Joyce English  | New Maximum    |
+--------+----------------+----------------+
5 rows in set (0.01 sec)
```

Employee ‘New Maximum’ is now the maximum earner for Job 2 and clearly the view was updated.

Ans – 8)

```
  GNU nano 6.2                                                    /t
create view Information
as
select e.Employee_id,Department_name,City,Country_name
from Employees as e
Join Departments as d on d.Department_id = e.Department_id
Join Locations as l on l.Location_id = d.Location_id
Join Countries as c on c.Country_id = l.Country_id;

select * from Information
 mysql> \e;
 Query OK, 0 rows affected (0.02 sec)

 +-------------+-----------------+---------+--------------+
 | Employee_id | Department_name | City    | Country_name |
 +-------------+-----------------+---------+--------------+
 |           1 | Research        | Noida   | India        |
 |           2 | Research        | Noida   | India        |
 |           3 | Administration  | Kyoto   | Japan        |
 |           4 | Administration  | Kyoto   | Japan        |
 |           5 | Research        | Noida   | India        |
 |           6 | Research        | Noida   | India        |
 |           7 | Research        | Noida   | India        |
 |           8 | Headquarters    | Gangtok | India        |
 |           9 | Headquarters    | Gangtok | India        |
 |          10 | Research        | Noida   | India        |
 |          11 | Administration  | Kyoto   | Japan        |
 +-------------+-----------------+---------+--------------+
 11 rows in set (0.03 sec)
```

Ans – 9)

```
  GNU nano 6.2                                                    /tmp/sqlhjWNql
# First we disable ONLY_FULL_GROUP_BY for ease of implementation
set GLOBAL sql_mode = (select replace(@@sql_mode,'ONLY_FULL_GROUP_BY',''));

select concat(m.First_name," ",m.Last_name) as Manager_name,m.Salary
from Employees as e
Join Employees as m on m.Employee_id = e.Manager_id
Group by e.Manager_id
Having COUNT(*) > 5;
```

```
mysql> \e;
Query OK, 0 rows affected (0.00 sec)

+---------------+--------+
| Manager_name | Salary |
+---------------+--------+
| John Smith    |  30000 |
+---------------+--------+
1 row in set (0.01 sec)
```

Ans – 10)

```
  GNU nano 6.2                                                    /tmp/sql2whipr
# First we disable ONLY_FULL_GROUP_BY for ease of implementation
set GLOBAL sql_mode = (select replace(@@sql_mode,'ONLY_FULL_GROUP_BY',''));

select concat(e1.First_name," ",e1.Last_name) as Name,e1.Salary
from Employees as e1
Left Join Employees as e2 on (e1.Department_id = 10 and e2.Department_id = 30 and e1.Salary >= e2.Salary)
where e2.Salary is NOT NULL
Group by e1.Employee_id
```

```
mysql> \e;
Query OK, 0 rows affected (0.00 sec)

+------------+--------+
| Name       | Salary |
+------------+--------+
| Ding Liren |  60000 |
| Frank Wong |  40000 |
+------------+--------+
2 rows in set (0.00 sec)
```

Ans – 11)

```
  GNU nano 6.2                                        /tmp/sqlrLPzfl
Create view Salaries
as
select concat(First_name," ",Last_name) as Name,Job_Title,Commission_pct,Salary * 12 as Annual_Salary,
Salary*12*(1+Commission_pct/100) as Annual_Salary_including_Commission

from Employees as e
Join Jobs as j on j.Job_id = e.Job_id
where Department_id = 20;

# Check view
select * from Salaries;

# Update Employees Salary by 10%
update Employees
Set Salary = Salary * 1.1;

select * from Salaries;

# Increase Commission of Clerks by 10%
update Employees as e
Join Jobs as j on j.Job_id = e.Job_id
Set Commission_pct = Commission_pct * (1+0.1*(Job_Title = 'Clerk'));

select * from Salaries;
```

```
mysql> \e;
Query OK, 0 rows affected (0.02 sec)

+---------------+------------+----------------+---------------+-----------------------------------+
| Name          | Job_Title  | Commission_pct | Annual_Salary | Annual_Salary_including_Commission |
+---------------+------------+----------------+---------------+-----------------------------------+
| Alicia Zelaya | Blacksmith |              5 |        300000 |                        315000.0000 |
| Jenni Wall    | Managerial |              7 |        516000 |                        552120.0000 |
| New Maximum   | Clerk      |             10 |       1140000 |                       1254000.0000 |
+---------------+------------+----------------+---------------+-----------------------------------+
3 rows in set (0.02 sec)

Query OK, 11 rows affected (0.03 sec)
Rows matched: 11  Changed: 11  Warnings: 0

+---------------+------------+----------------+---------------+-----------------------------------+
| Name          | Job_Title  | Commission_pct | Annual_Salary | Annual_Salary_including_Commission |
+---------------+------------+----------------+---------------+-----------------------------------+
| Alicia Zelaya | Blacksmith |              5 |        330000 |                        346500.0000 |
| Jenni Wall    | Managerial |              7 |        567600 |                        607332.0000 |
| New Maximum   | Clerk      |             10 |       1254000 |                       1379400.0000 |
+---------------+------------+----------------+---------------+-----------------------------------+
3 rows in set (0.03 sec)

Query OK, 3 rows affected (0.03 sec)
Rows matched: 11  Changed: 3  Warnings: 0

+---------------+------------+----------------+---------------+-----------------------------------+
| Name          | Job_Title  | Commission_pct | Annual_Salary | Annual_Salary_including_Commission |
+---------------+------------+----------------+---------------+-----------------------------------+
| Alicia Zelaya | Blacksmith |              5 |        330000 |                        346500.0000 |
| Jenni Wall    | Managerial |              7 |        567600 |                        607332.0000 |
| New Maximum   | Clerk      |             11 |       1254000 |                       1391940.0000 |
+---------------+------------+----------------+---------------+-----------------------------------+
3 rows in set (0.03 sec)
```